

6.2 Linear Combination



Core skills you'll master in this section:

- ▶ $\mathbf{a}_1 + k\mathbf{a}_2$: the line through the endpoint of $\mathbf{a}_1$, parallel to $\mathbf{a}_2$.
- ▶ $\mathbf{a}_1 + k_2\mathbf{a}_2 + k_3\mathbf{a}_3$: the plane through the endpoint of $\mathbf{a}_1$, parallel to $\mathbf{a}_2$ and $\mathbf{a}_3$.
- ▶ $k\mathbf{a}_1 + (1 - k)\mathbf{a}_2$: the line through the endpoints of $\mathbf{a}_1$ and $\mathbf{a}_2$; linear interpolation.
- ▶ $k\mathbf{1} + (1 - k)\mathbf{a}$: the line through the endpoints of $\mathbf{1}$ and $\mathbf{a}$; linear interpolation.
- ▶ $k_1\mathbf{a}_1 + k_2\mathbf{a}_2 + (1 - k_1 - k_2)\mathbf{a}_3$: the plane through the endpoints of $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$.

I'm sure you're already quite familiar with linear combinations; as a reminder, a linear combination is a new vector constructed from the weighted sum of a set of vectors and their corresponding scalars (coefficients). In this section, let's look at a few interesting linear combinations that occur in RGB space, and how they connect algebra, linear algebra, and geometry. There are no exercises in this section, so you can read through it at ease.

$$\mathbf{a}_1 + k\mathbf{a}_2$$

Given that $\mathbf{a}_2$ is an n -dimensional column vector, if k is any real number, $k\mathbf{a}_2$ represents the one-dimensional space $\text{span}(\mathbf{a}_2)$. This vector space clearly passes through the origin, that is, $\mathbf{0}$.

$\mathbf{a}_1 + k\mathbf{a}_2$ is equivalent to translating $k\mathbf{a}_2$ by $\mathbf{a}_1$, and the result is an affine space.

Let's turn to the RGB color space, and see what $\mathbf{a}_1 + k\mathbf{a}_2$ represents.

As shown in Figure 2 (b), $k_2\mathbf{e}_2$ represents "dark green," which includes pure green and pure black. From a vector-space perspective, $\text{span}(\mathbf{e}_2)$ is one-dimensional.

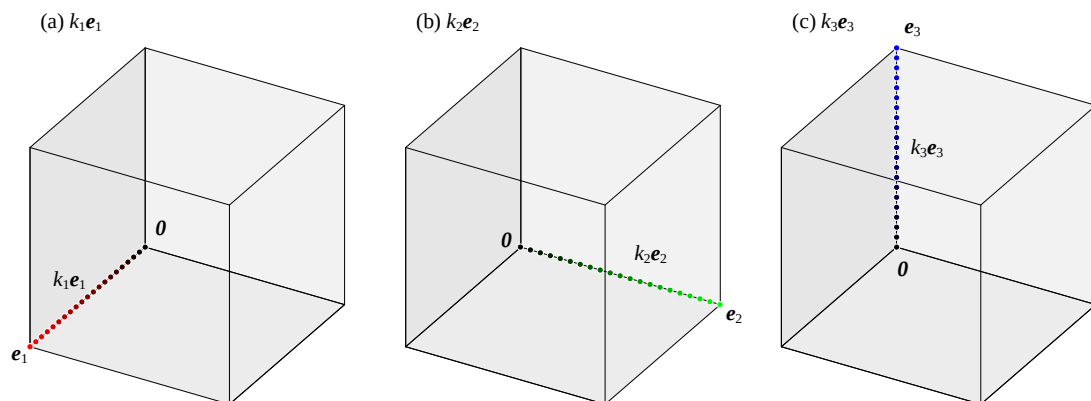


Figure 1. In the RGB Color Space, $k\mathbf{a}_2$

As shown in Figure 2 (a.1), after translating by $\mathbf{e}_1$, we get $\mathbf{e}_1 + k_2\mathbf{e}_2$, and the color changes from red to yellow; it does not include the pure black vector, that is, it does not pass through the origin.

? Please analyze Figure 2's remaining subplots on your own.

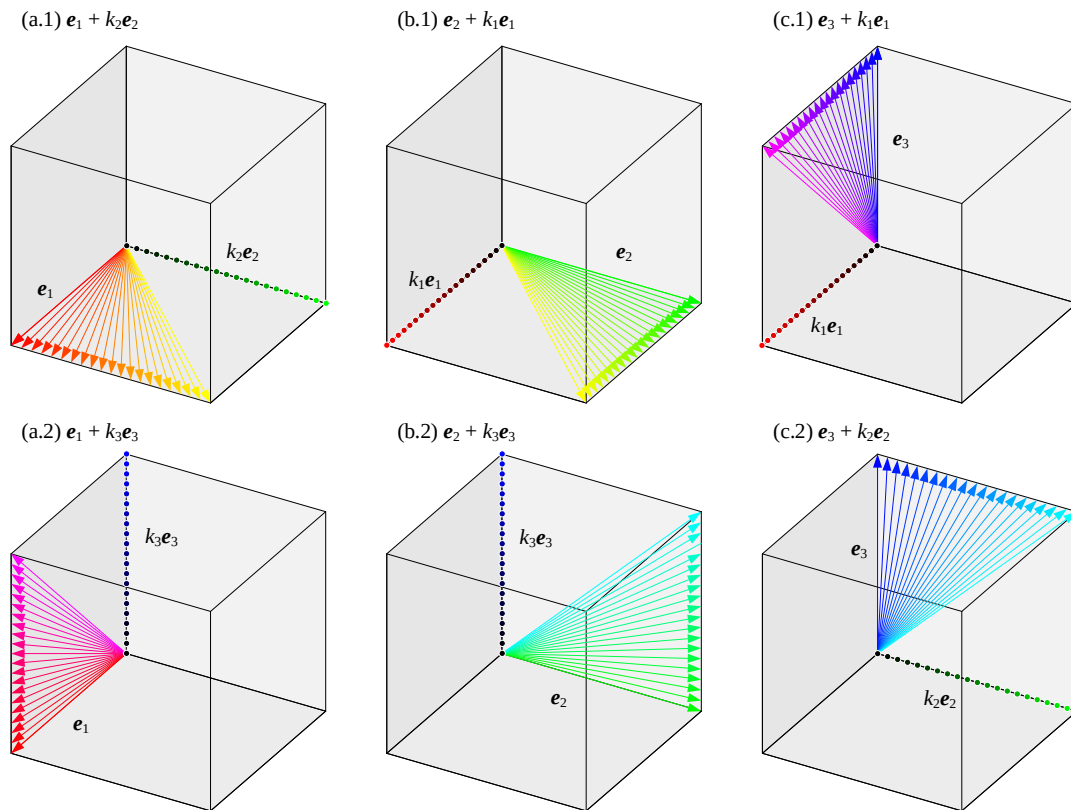


Figure 2. In the RGB Color Space, $\mathbf{a}_1 + k\mathbf{a}_2$

$$\mathbf{a}_1 + k_2\mathbf{a}_2 + k_3\mathbf{a}_3$$

Let's look at a second example of a linear combination — $\mathbf{a}_1 + k_2\mathbf{a}_2 + k_3\mathbf{a}_3$.

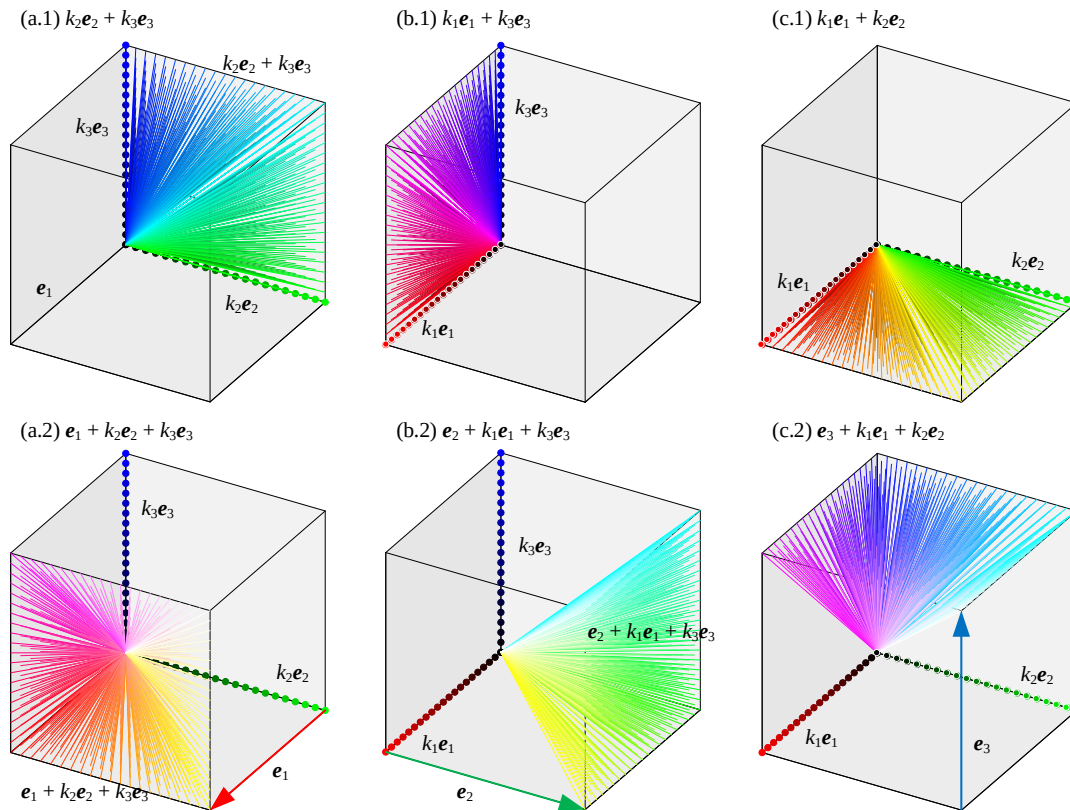
When k_2 and k_3 take any real values, $k_2\mathbf{a}_2 + k_3\mathbf{a}_3$ spans a plane, that is, $\text{span}(\mathbf{a}_2, \mathbf{a}_3)$. This vector space is two-dimensional.

Then, $\mathbf{a}_1 + (k_2\mathbf{a}_2 + k_3\mathbf{a}_3)$ represents translating $\text{span}(\mathbf{a}_2, \mathbf{a}_3)$ by $\mathbf{a}_1$, and the result is an affine space.

As shown in Figure 3 (a.1), $k_2\mathbf{e}_2 + k_3\mathbf{e}_3$ is equivalent to the green and blue vectors spanning a "blue-green" plane; this plane contains the black vector, that is, the zero vector $\mathbf{0}$.

Translating the plane $k_2\mathbf{e}_2 + k_3\mathbf{e}_3$ by $\mathbf{e}_1$, we get Figure 3 (a.2).

? Please analyze Figure 3's remaining subplots on your own.

Figure 3. In the RGB Color Space, $\mathbf{a}_1 + k_2\mathbf{a}_2 + k_3\mathbf{a}_3$

$$k\mathbf{a}_1 + (1 - k)\mathbf{a}_2$$

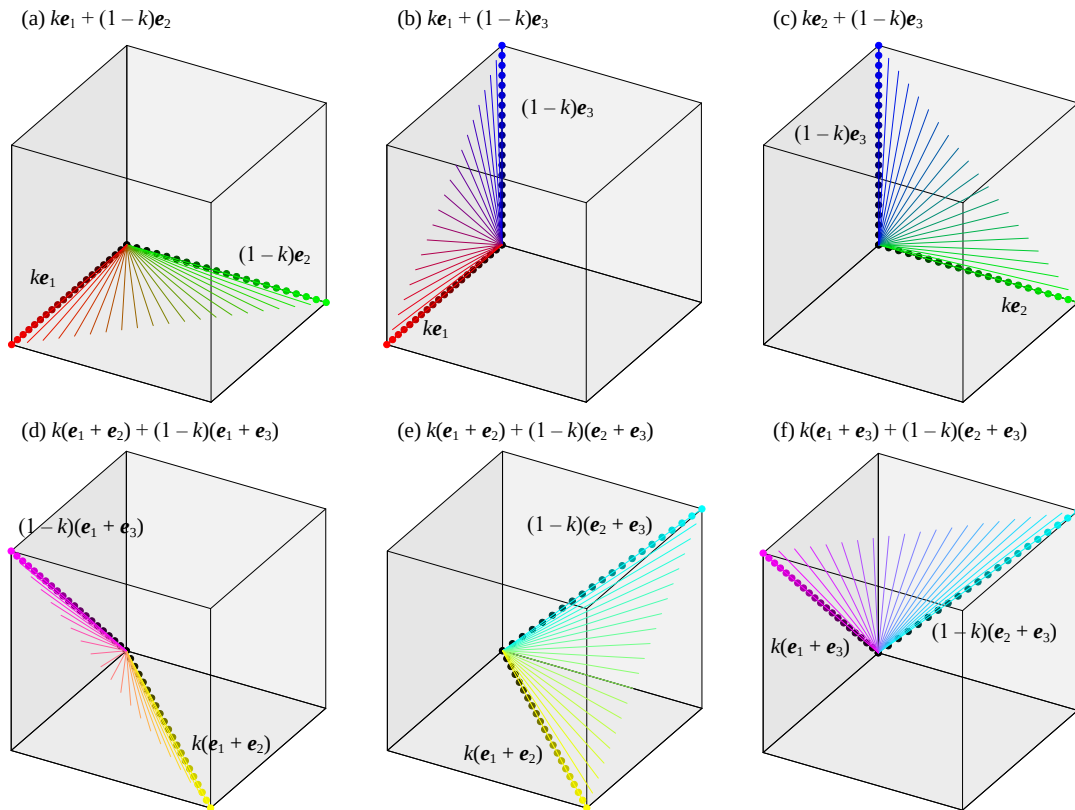
When k takes any real value, $k\mathbf{a}_1 + (1 - k)\mathbf{a}_2$ gives the line connecting the endpoints of the vectors $\mathbf{a}_1$ and $\mathbf{a}_2$, which is equivalent to linear interpolation.

From a geometric perspective, linear interpolation is simply drawing a line between two points and finding the corresponding intermediate point on that line.

Figure 4 (a) shows linear interpolation between red and green; this line segment is the connection between the endpoint of the red vector and the endpoint of the green vector.



Please analyze Figure 4's remaining subplots on your own.

Figure 4. In the RGB Color Space, $k\mathbf{a}_1 + (1 - k)\mathbf{a}_2$

$$k\mathbf{1} + (1 - k)\mathbf{a}$$

Earlier in this book, we used darkening a given color to explain scalar multiplication of vectors, and left a question open — how do we lighten a given color?

The linear combination $k\mathbf{1} + (1 - k)\mathbf{a}$ can accomplish this. The all-ones column vector $\mathbf{1}$ represents white, and $\mathbf{a}$ can be any given color vector. The scalar k ranges over $[0, 1]$; the closer k is to 1, the closer the color gets to white, and the lighter it becomes; the closer k is to 0, the closer the color gets to the original color.

As shown in Figure 5, for the linear combination $k\mathbf{1} + (1 - k)\mathbf{e}_1$, when $k = 0$, the color is red; when $k = 1$, the color is white.



Please analyze Figure 5's remaining subplots on your own.

Figure 6 shows two creative-coding examples of lightening colors in the RGB color space.

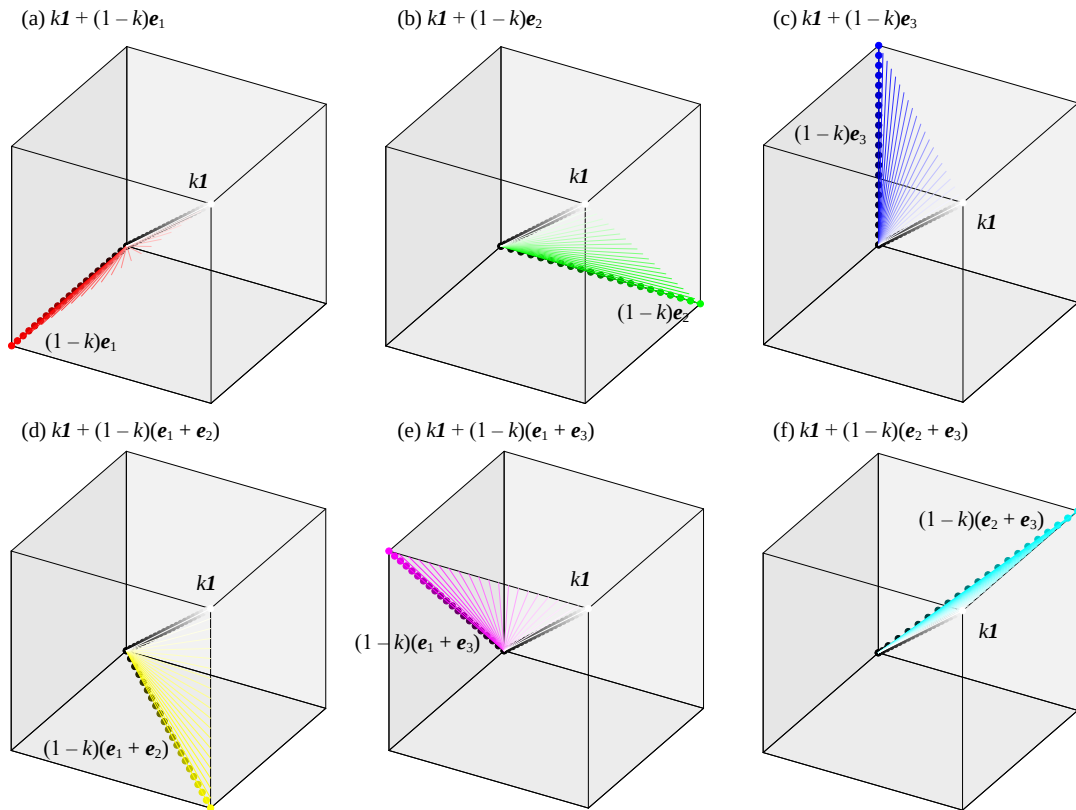
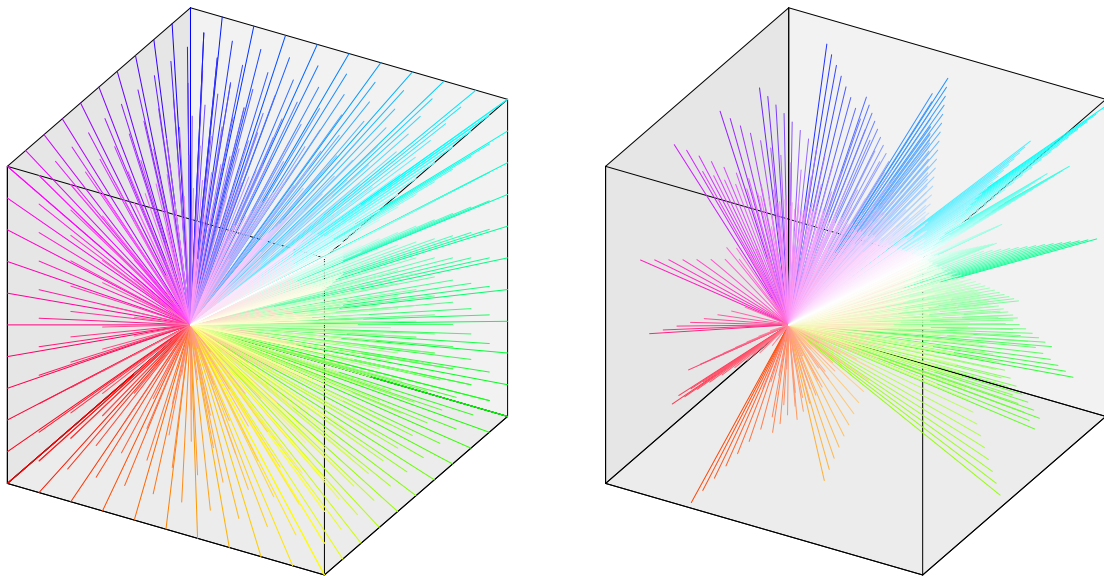
Figure 5. In the RGB Color Space, $k\mathbf{1} + (1-k)\mathbf{a}$ 

Figure 6. Creative Coding for Lightening Colors in the RGB Color Space

$$k_1\mathbf{a}_1 + k_2\mathbf{a}_2 + (1 - k_1 - k_2)\mathbf{a}_3$$

The linear combination $k_1\mathbf{a}_1 + k_2\mathbf{a}_2 + (1 - k_1 - k_2)\mathbf{a}_3$ spans a plane; this plane passes through the endpoints of the three vectors $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$.

Figure 7 (a) shows the plane formed by the red, green, and blue vectors; Figure 7 (b) shows the plane formed by the yellow, magenta, and cyan vectors.

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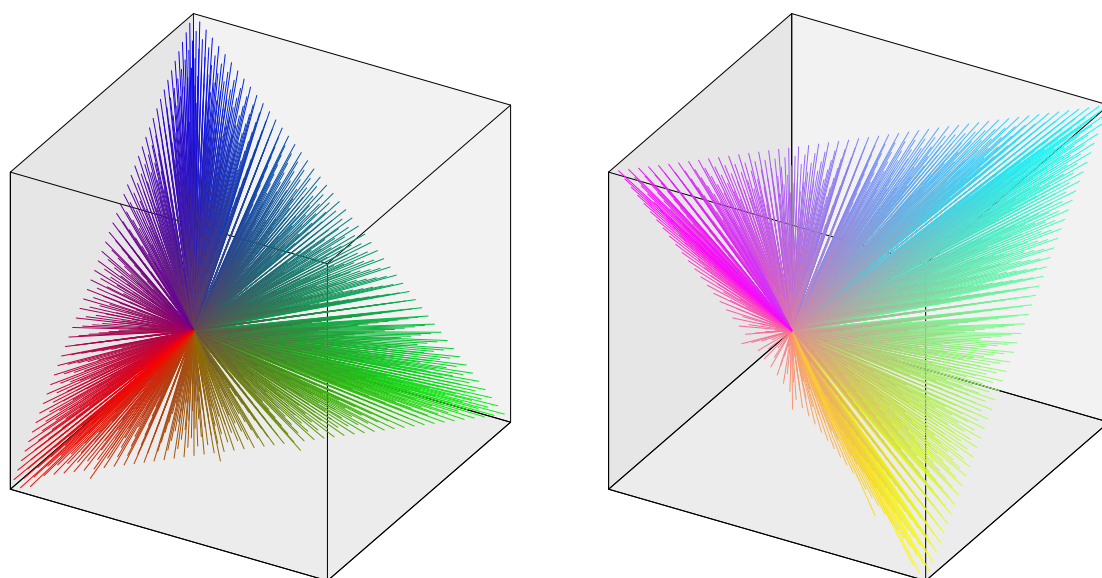


Figure 7. A Plane Spanned in the RGB Color Space